

EDUCATION

Northeastern University - Boston, MA

May 2025

M.S. in Robotics, Concentration in Mechanical Engineering

GPA: 3.92

B.S. in Mechanical Engineering, Minors in Mathematics & Religious Studies

GPA: 3.83

Coursework: Control Systems, Mechatronic Systems, Robot Science and Systems, Mobile Robotics, Mechanical Design, Mechanics of Materials, Computation & Design, Heat Transfer, Fluid Dynamics, Dynamics

Honors and Awards: Dean's List (All Semesters), Dean's Connections Scholarship Recipient, Ukrainian Aviation and Maritime Eagle Scout

Activities: Students for the Exploration and Development of Space (SEDS), VP of Theme Park Engineering Club, American Society of Mechanical Engineers

SKILLS

Software: SolidWorks CSWA (Surfacing, FEA, Solid Modeling), OnShape, Fusion 360, AutoCAD, 3D Printing (FDM/SLA), MS Office Suite, Arduino, MATLAB (Simulink + Simscape), Python (basic), C++ (basic)

Equipment: Soldering, Laser Cutting, Metal Stamping, Standard Shop Tools (Lathe, Mill, etc.)

PROFESSIONAL EXPERIENCE

Rise Robotics | *Mechanical Engineering Intern* | Somerville, MA

Feb - June 2026

- Diagnosed and repaired failures across 5 belt driven linear actuators (noise, scraping, breakage) and added a spacer shim to eliminate rattle and informed design recommendations for future cylinder revisions
- Ran SolidWorks motion studies to isolate chain binding during platform closure and iterated 3 prototypes to a linkage driven catch guiding the chain through a curved channel in a tight packaging envelope
- Identified a safety gap in a high pressure test area: designed and installed a magnetic reed switch interlock to auto terminate testing when the test cell door opens

Hasbro | *Animatronics Engineering Co-Op* | Pawtucket, RI

July - Dec. 2023

- Led electrical component selection, assembly, and testing of PCB's including components like resistors, headers, lights, and switches for toy animation with a focus on design for production and cost savings
- Designed dynamic toy elements in Solidworks (100+ hours) with gear trains, 4-bar linkages, and springs to incorporate movement in animated toys
- Machined parts for toys using standard shop tools such as a lathe, mill, drill press, and sander

Epicro Biosystems | *Mechanical Engineering Co-Op* | Cambridge, MA

July - Dec. 2022

- Designed a waterproofing test procedure by IPX4 Standards to subject units to water submersion and exposure at 50-150 kPa to dictate areas of improvement for product redesign
- Authored and tested a manufacturing procedure utilized for refurbishing over 60 Connected Hydration Devices to check PCB function and replace faulty housings and gaskets
- Optimized post-processing Python scripts for on-body unit tests, reducing analysis time by 96%

PROJECTS/ACTIVITIES

Crochet Machine, Masters Thesis Project

Sept. 2024 - April 2025

- Designed, built, and programmed an automated crochet machine that autonomously creates a chain stitch via a three-part system (crocheter, tensioner, holder) using 8020 aluminum framing, servo/DC motors, and 3D-printed PLA parts
- Implemented open-loop motion control for linear/rotational actuation via rack-and-pinion, plus motorized tension control for precise yarn feeding
- Developed a MATLAB-based pattern generator to model project dimensions and parametrically adjust chain length, hook size, and yarn type for custom outputs
- Ran motion capture experiments on human crochet motion to inform design kinematics

ErgPAL, Senior Year Capstone Project

May 2023 - May 2024

- Designed and prototyped a rowing machine accessory using a rotating eccentric mass to replicate in-boat impulse cues, helping athletes train solo for better synchronization
- Programmed an ESP-32 microcontroller to send PWM signals to a brushless DC motor with a custom control system, integrating ultrasonic sensors for positional accuracy
- Soldered the electrical system with a Spark MAX motor controller, emergency stop button, potentiometer for stroke rate adjustment, and LED status indicator
- Conducted iterative tests and analysis to optimize impulse forces and motor dynamics